INCH-POUND

MIL-M-28787/117A
AMENDMENT 1
28 April 1993

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE BDD

This amendment forms a part of Military Specification MIL-M-28787/117A(EC), dated 4 June 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 3

FIGURE 1, Circuit 2, Output at pin 34: Delete label "Q1" and substitute "Q2".

PAGE 4

TABLE I, Test conditions B1 and B2, Test type column: Delete "X" and substitute "S".

TABLE I, Test conditions D1-D4, Test type column: Delete "X" and substitute "S".

TABLE I, Test condition E1, Test type column: Delete "S" and substitute "X".

TABLE I, Test condition E2, Test type column: Delete "S" and substitute "X".

PAGE 5

TABLE I, Test conditions P1-P12, Test type column: Delete "S" and substitute "Q".

PAGE 6

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through R shall be performed in any order, with the following exceptions. Test conditions B, D1-D4, Q, and R are stress tests and shall be performed only at 25 ± 5°C. If test conditions B, D1-D4, Q, or R are part of a sequence of tests, the order of testing shall be as follows: B, D1-D4, Q, and R where applicable, followed by the remaining tests in the sequence."

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4.5.1: Delete and substitute the following:

"4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A, C, D5-D8, and E through N shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3."

4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests A, C, D5-D8, and E through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests A, C, D5-D8, and E through P shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I."

PAGE 18

TABLE VI, Test condition F20, Parameter name: Delete "Q2" and substitute "$\overline{Q}$2".

PAGE 20

TABLE VI, Test condition F60, Parameter name: Delete "Q2" and substitute "$\overline{Q}$2".

PAGE 25

FIGURE 9: Delete and substitute new figure 9 as printed on pages 4 and 5 of this amendment.

PAGE 26

FIGURE 10: Delete and substitute new figure 10 as printed on pages 6 and 7 of this amendment.

PAGE 35

FIGURE 14, Note 7: Delete and substitute the following:

"7. For test conditions N7 and N8, connect item J01 to R2; connect item Y04 probe to module pin 1; and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5V ± 3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9V ± 5%."

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PAGE 39

FIGURE 17: Delete and substitute new figure 17 as printed on page 8 of this amendment.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0131)

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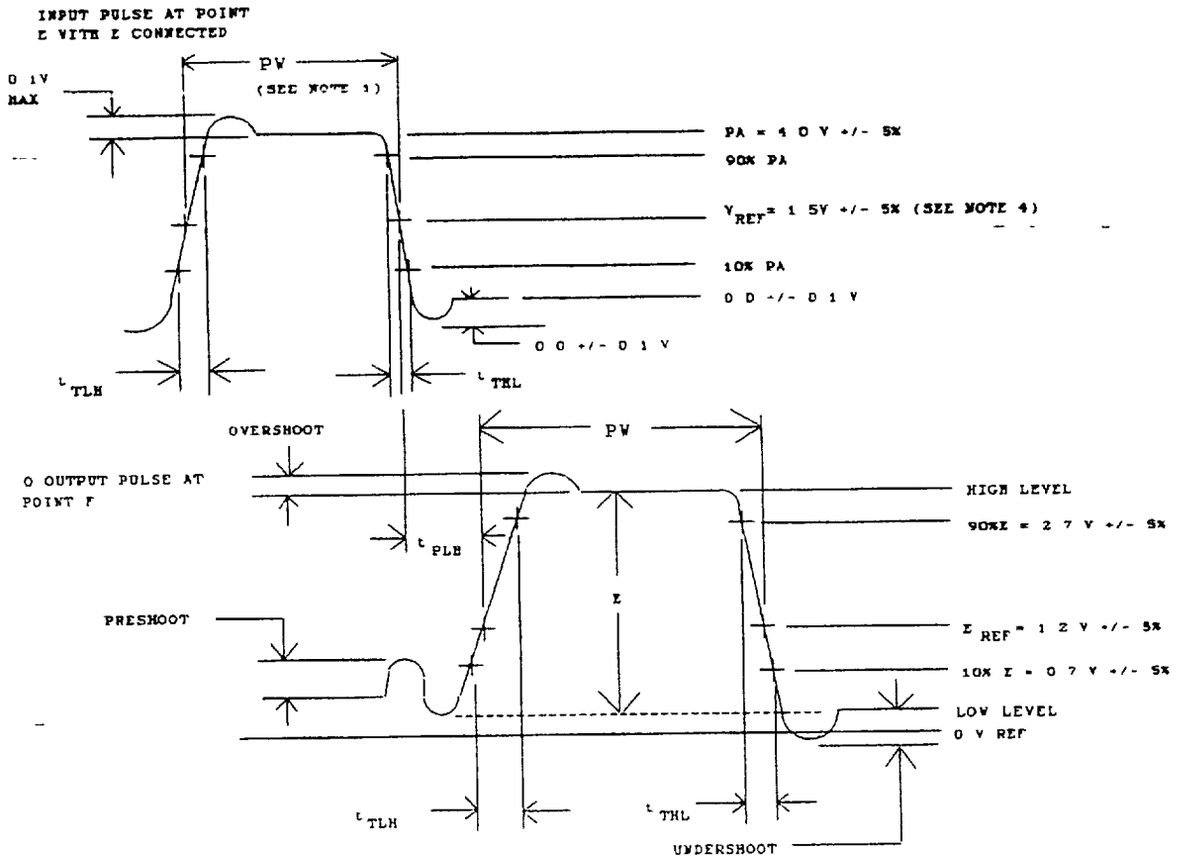


FIGURE 9. Output pulse width (Q and $\overline{Q}$), transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot, and preshoot), and input sensitivity (trigger) and immunity test circuit waveforms for tables VI through X, and XII.

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NOTES:

1. Input pulse parameters:
   PA = 4.0V ± 5%.
   $t_{TLH} = t_{THL}$ = 25 ± 5 ns.
   PW = 125 ns ± 10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60, G1-G8, H1-H4, J1-J8, and K1-K12.
   = 100 µs ± 10% for test conditions F9-F12, F17-F28 and F33-F56.
   PRP = 300 µs ± 10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60, G1-G8, H1-H4, J1-J8, and K1-K12.
   = 200 µs ± 10% for test conditions F9-F12, F17-F28 and F33-F56.

2. For test conditions G1-G8, H1-H4, J1-J8 and K1-K12, connect item Y01-2 as close as physically possible to the module pin.

3. Several perturbations which may occur on the output waveform are shown (preshoot, overshoot, undershoot). Perturbations may occur at other locations on the output waveforms, but they must conform to the following requirements: No negative perturbations on the high level or positive perturbations on the low level shall exceed 0.5V amplitude.

4. Input pulse width is measured at 50% PA for test conditions M1-M12.

FIGURE 9. Output pulse width (Q and $\overline{Q}$), transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot, and preshoot), and input sensitivity (trigger) and immunity test circuit waveforms for tables VI through X, and XII - Continued.

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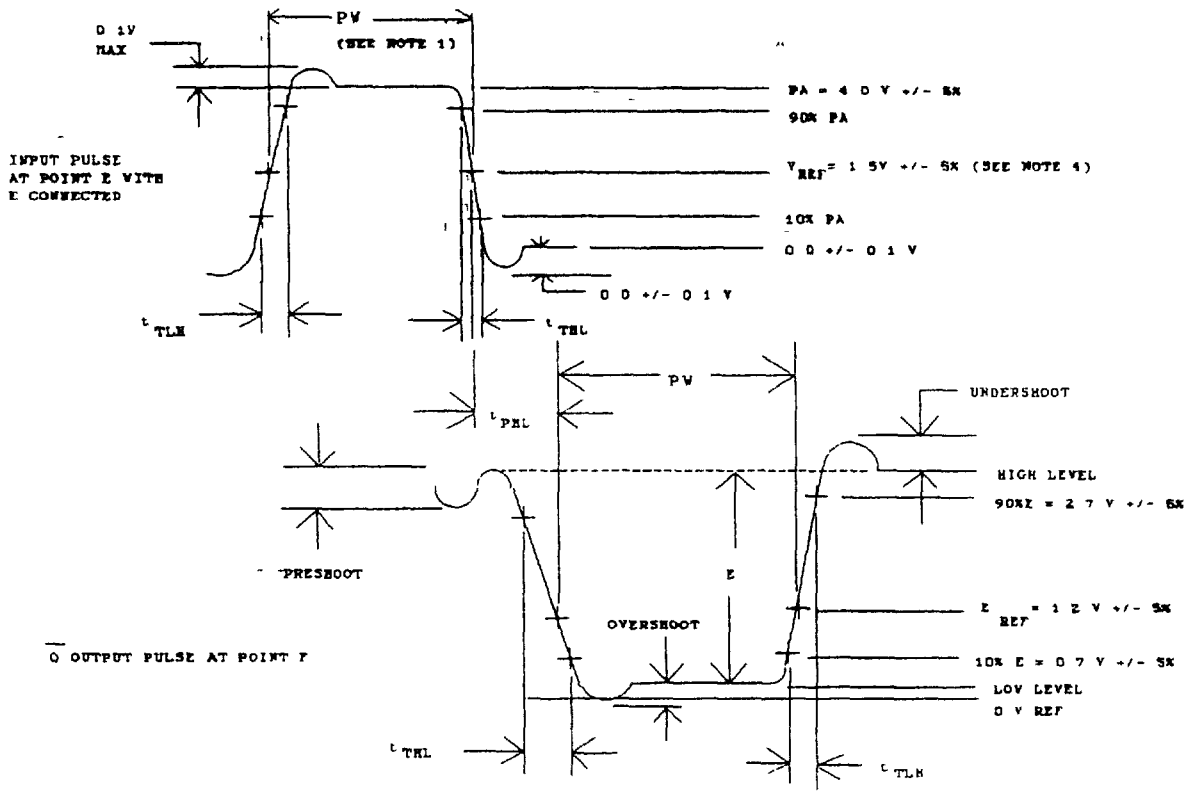


FIGURE 10. Output pulse width (Q and $\overline{Q}$), transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot, and preshoot), and input sensitivity (trigger) and immunity test circuit waveforms for tables VI through X, and XII.

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NOTES:

1. Input pulse parameters:
   PA = 4.0V ± 5%.
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   PW = 125 ns ± 10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60, G1-G8, H1-H4, J1-J8, and K1-K12.
   = 100 µs ± 10% for test conditions F9-F12, F17-F28 and F33-F56.
   PRP = 300 µs ± 10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60, G1-G8, H1-H4, J1-J8, and K1-K12.
   = 200 µs ± 10% for test conditions F9-F12, F17-F28 and F33-F56.

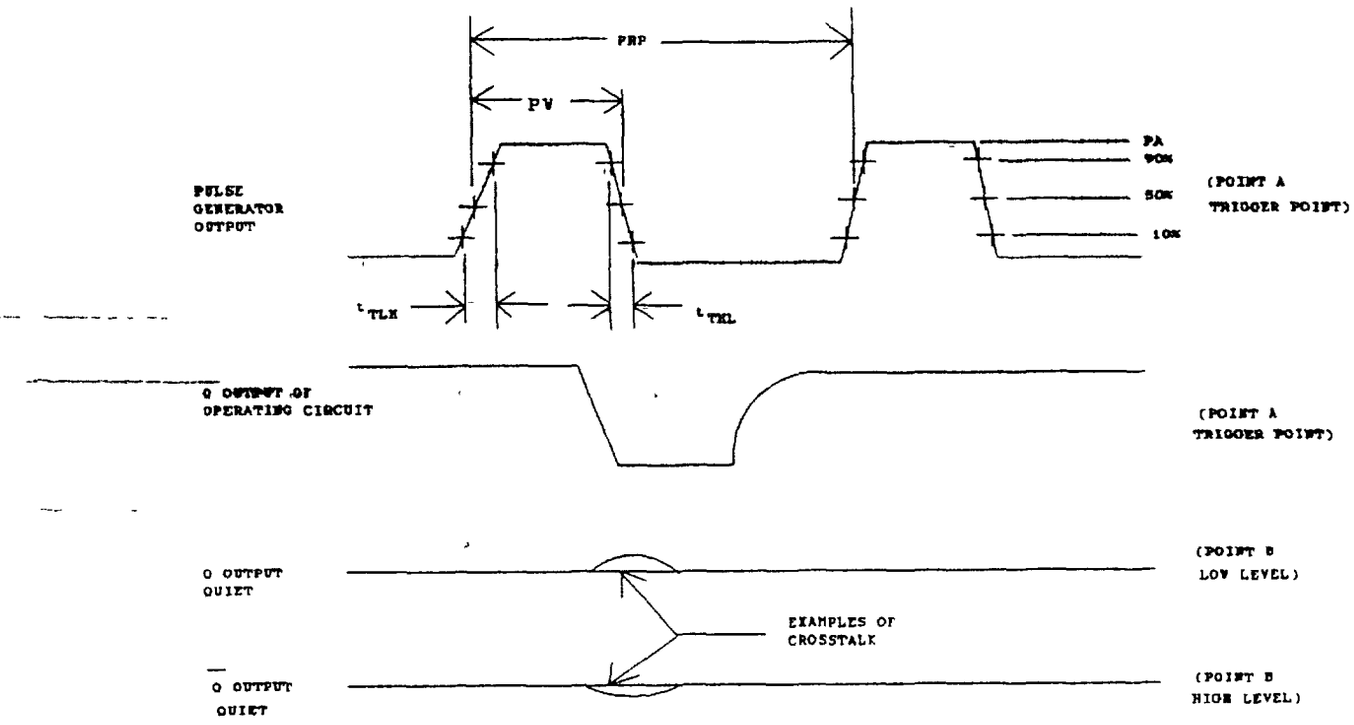
2. For test conditions G1-G8, H1-H4, J1-J8 and K1-K12, connect item Y01-2 as close as physically possible to the module pin.

3. Several perturbations which may occur on the output waveform are shown (preshoot, overshoot, undershoot). Perturbations may occur at other locations on the output waveforms, but they must conform to the following requirements: No negative perturbations on the high level or positive perturbations on the low level shall exceed 0.5V amplitude.

4. Input pulse width is measured at 50% PA for test conditions M1-M12.

FIGURE 10. Output pulse width (Q and $\overline{Q}$), transition time (t_{THL} and t_{TLH}), propagation delay time (t_{PLH} and t_{PHL}), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot, and preshoot), and input sensitivity (trigger) and immunity test circuit waveforms for tables VI through X, and XII - Continued.

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NOTES:

1. Adjust the output of item J01 to obtain the following input pulse:
   PA = 4 V ± 5%.
   PW (2 V level) = 125 ns ± 10%.
   $t_{TLH} = t_{THL}$ = 10 ns ± 10%.
   PRP = 3 μs ± 10% (use items D03 and Y04 for this measurement).
2. To examine the output waveforms under test, item D05 shall be internally triggered using the signal on channel A (item Y01-1).
3. For each test condition, the maximum pulse width detected at point B, measured at 1.0 V above the low level on the Q output and at 2.0 V below the high level on the Q output, shall not exceed the limits in table I.

FIGURE 17. Crosstalk test circuit notes and waveforms for table XIV.

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